



Vortex - Ares ENSEA



Identity Card

Country:	France
City:	Cergy
1st C'space:	2024
Previous Launch:	2025 Les Crackitos
Current project:	2026 Vortex

About Us

Ares is the robotics association of ENSEA (École nationale supérieure de l'électronique et de ses applications).

The Vortex team built on last year's expertise and created a wind turbine with conter rotative drone propellers to test the rotation and chute deployment mechanism.

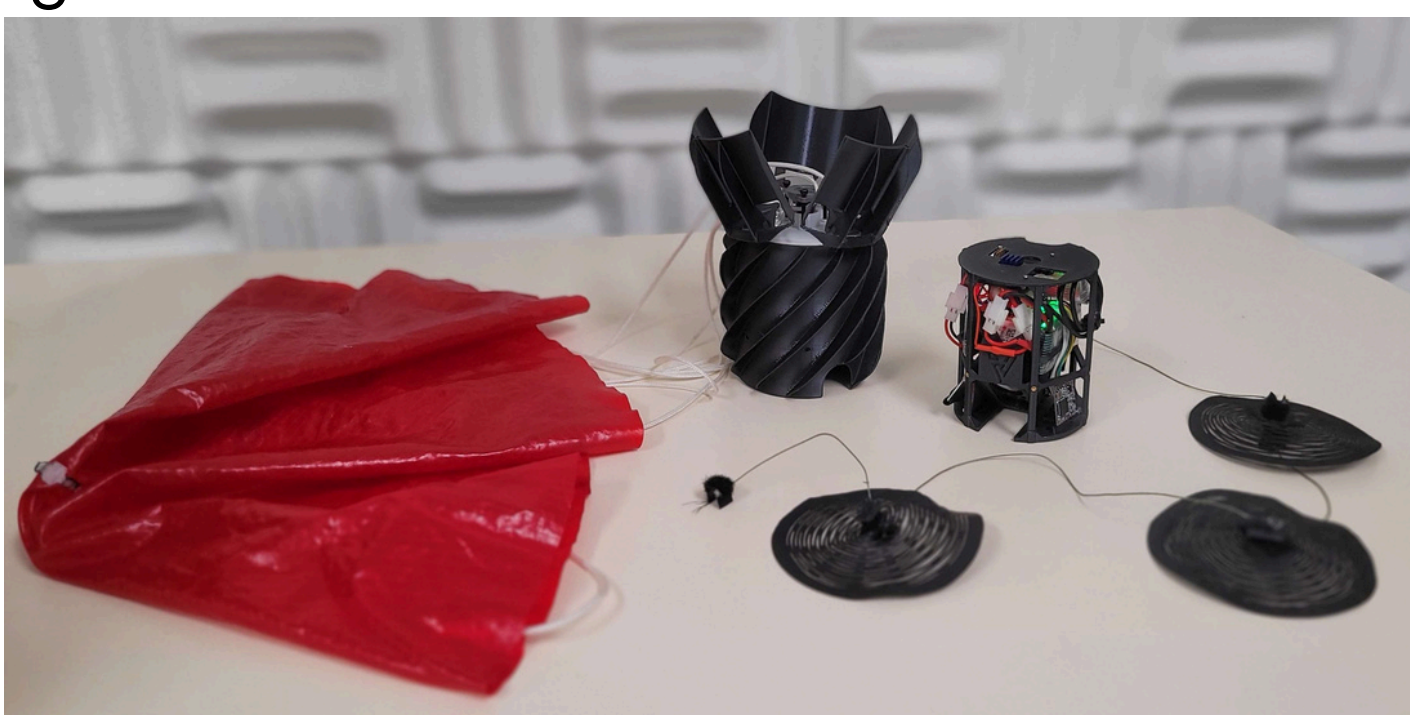


Team Member	Role
Alexei DOUILLARD	3D Modeling, PCB, Satellite integration, Head of project
Juan Pablo BARONA CIFUENTES	Embedded systems, firmware
Ted KOYAZANDE	Ground Station, Communications
Amr TAOUIS	PCB Design and testing
Abdelmoughit HAJJI LAAMOURI	Code



Cansat 2026 missions

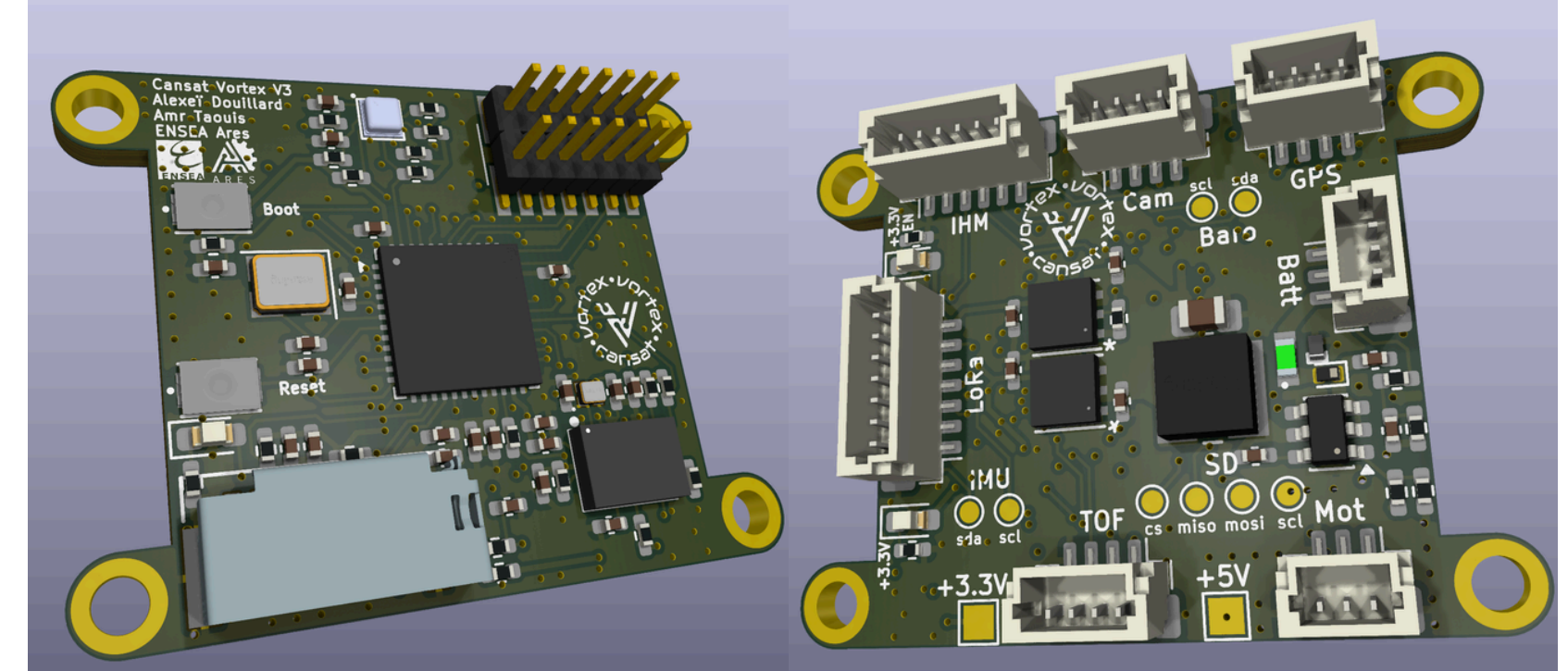
- **Downsizing** : Integrating the core system within the volume of a 33cl soda can
- **Deployment of a fully contained parachute deployed between 75 and 90m** : Using an extractor parachute (drogue chute) inspired by kirigami calibrated for the 15m window.
- **Ground Mapping** : Generating a 3D map of the terrain through rotation using a LiDAR, GPS, barometer, and an inertial measurement unit.
- **Onboard Camera** : Recording the landing using a drone camera.
- **Bonus mission** : telemetry transmission to a ground station



Vortex Cansat C'SPACE 2026

• Hardware : Electronics

Design of a 6-layer PCB integrating an STM32 microcontroller, SD card, barometer, and 9 axis IMU, with dedicated connectivity for GPS, LoRa, camera, and LiDAR modules.

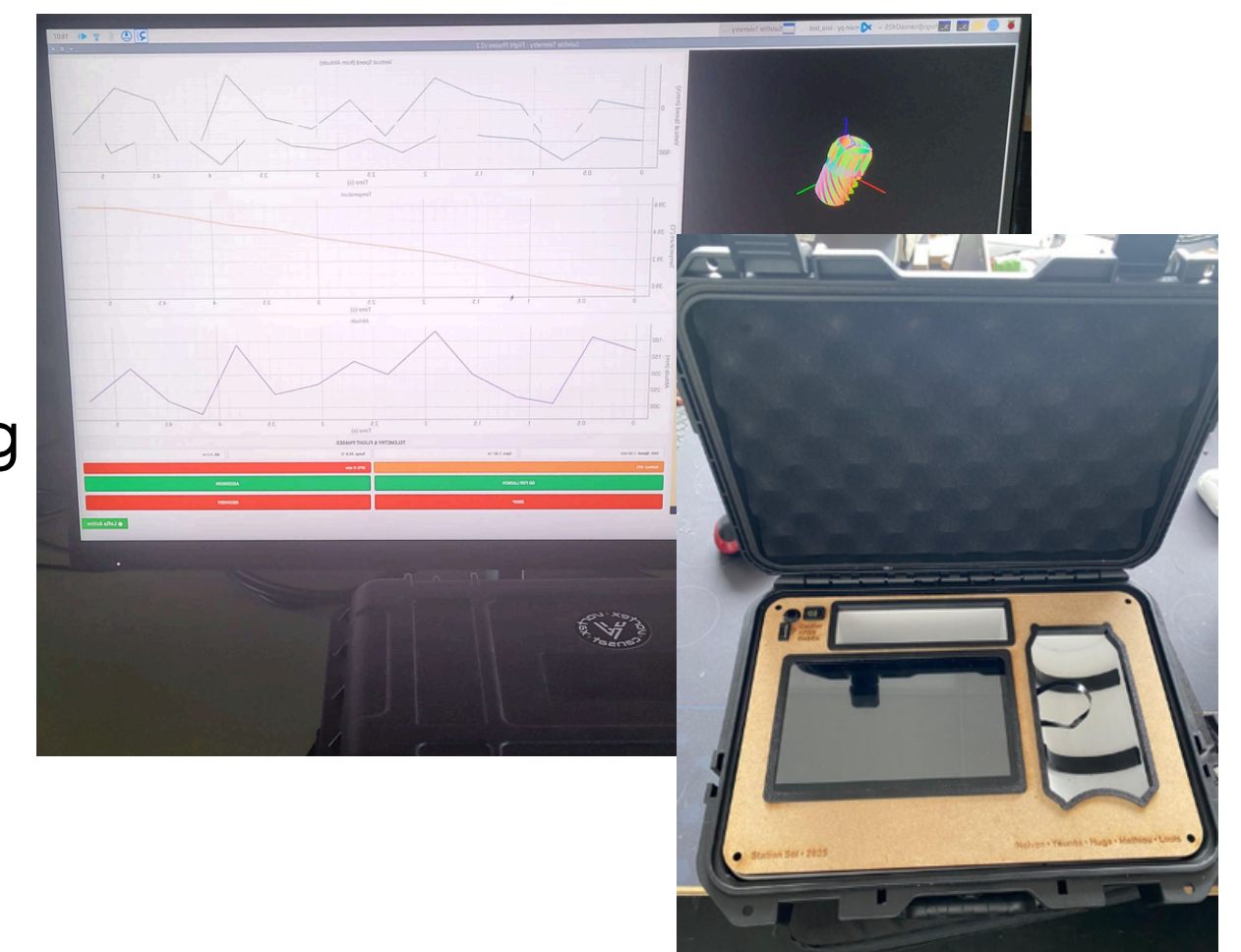


• Software & Telemetry

Development of an embedded software architecture using FreeRTOS (C programming language). It manages sensor fusion, data logging to the SD card, and real-time flight data transmission via a LoRa module.

• Ground Station & HMI

Creation of an autonomous ground station powered by a Raspberry Pi 5. It features a dashboard with real-time 3D orientation modeling of the CanSat, GPS tracking on a map, and live telemetry graphs. We also built a plug in Human Machine Interface for the preflight checks.



• Mechanics & Aerodynamics

The 3D design of the satellite has a helical body engineered to induce rotation during descent. It enables the lidar to scan the floor and create a spiral topographical map. The electronics is secured in an internal skeleton. To deploy the main Ø45cm parachute, laser cut kirigami drogue parachutes pull on a trigger. To ensure deployment at a precise speed we tuned the trigger to adjust friction and chained drogues chutes to get enough force.

